In [1]:

```python
import pandas as pd
df1=pd.read_csv('df1.csv')
```

In [2]:

```python
df1
```

Out[2]:

	Unnamed: 0	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene
0	30	Ahmedabad	2015-01-02	135.99	123.974957	43.48	42.08	84.57	20.655194	43.48	75.23	102.70	0.40	0.04
1	31	Ahmedabad	2015-02-02	178.33	123.974957	54.56	35.31	72.80	20.655194	54.56	55.04	107.38	0.46	0.06
2	32	Ahmedabad	2015-03-02	139.70	123.974957	30.61	28.40	56.73	20.655194	30.61	33.79	73.60	0.17	0.03
3	33	Ahmedabad	2015-04-02	80.65	123.974957	2.37	22.83	24.00	20.655194	2.37	25.73	47.30	0.00	0.00
4	34	Ahmedabad	2015-05-02	58.36	123.974957	2.60	21.39	23.31	20.655194	2.60	32.66	53.54	0.00	0.00
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3528	9283	Visakhapatnam	2020-08-06	31.14	125.150000	6.55	41.56	27.26	9.710000	0.64	6.61	36.65	3.47	9.76
3529	9284	Visakhapatnam	2020-09-06	31.84	85.410000	4.69	50.11	30.35	10.010000	0.88	5.06	33.86	4.07	9.24
3530	9285	Visakhapatnam	2020-10-06	32.55	79.280000	4.78	44.59	27.38	9.340000	0.69	5.90	35.06	2.96	7.89
3531	9286	Visakhapatnam	2020-11-06	37.71	98.530000	9.97	52.10	35.79	9.910000	0.70	5.59	26.59	5.29	10.38
3532	9287	Visakhapatnam	2020-12-06	37.67	76.270000	12.17	53.59	35.52	9.850000	0.97	5.74	17.35	4.77	18.89

3533 rows × 17 columns

In [3]:

```python
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error

print("Columns in df1:", df1.columns)

X = df1[['NO2', 'SO2', 'PM10']]
y = df1['PM2.5']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

y_pred = lr_model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
print(f"Mean Squared Error for Linear Regression: {mse}")
```

```
Columns in df1: Index(['Unnamed: 0', 'City', 'Date', 'PM2.5', 'PM10', 'NO', 'NO2', 'NOx',
       'NH3', 'CO', 'SO2', 'O3', 'Benzene', 'Toluene', 'Xylene', 'AQI',
       'AQI_Bucket'],
      dtype='object')
Mean Squared Error for Linear Regression: 2383.588995869094
```

In [4]:

```python
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import StandardScaler

print("First 5 rows:\n", df1.head())
print("Available columns:\n", df1.columns)

df1 = df1.dropna()

X = df1[['NO2', 'SO2', 'PM10']]
y = df1['PM2.5']

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)
```

```python
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

y_pred = lr_model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"R² Score: {r2:.2f}")

print("\nModel Coefficients:")
for feature, coef in zip(['NO2', 'SO2', 'PM10'], lr_model.coef_):
    print(f"{feature}: {coef:.4f}")

result_df = pd.DataFrame({
    'Actual': y_test,
    'Predicted': y_pred
})
print("\nPrediction Results Sample:\n", result_df.head())
```

```
First 5 rows:
    Unnamed: 0       City        Date   PM2.5        PM10     NO    NO2    NOx  \
0           30  Ahmedabad  2015-01-02  135.99  123.974957  43.48  42.08  84.57
1           31  Ahmedabad  2015-02-02  178.33  123.974957  54.56  35.31  72.80
2           32  Ahmedabad  2015-03-02  139.70  123.974957  30.61  28.40  56.73
3           33  Ahmedabad  2015-04-02   80.65  123.974957   2.37  22.83  24.00
4           34  Ahmedabad  2015-05-02   58.36  123.974957   2.60  21.39  23.31

         NH3     CO    SO2      O3  Benzene  Toluene  Xylene    AQI AQI_Bucket
0  20.655194  43.48  75.23  102.70     0.40     0.04   25.87  782.0     Severe
1  20.655194  54.56  55.04  107.38     0.46     0.06   35.61  914.0     Severe
2  20.655194  30.61  33.79   73.60     0.17     0.03   11.87  660.0     Severe
3  20.655194   2.37  25.73   47.30     0.00     0.00    0.00  294.0       Poor
4  20.655194   2.60  32.66   53.54     0.00     0.00    0.00  149.0   Moderate
Available columns:
 Index(['Unnamed: 0', 'City', 'Date', 'PM2.5', 'PM10', 'NO', 'NO2', 'NOx',
       'NH3', 'CO', 'SO2', 'O3', 'Benzene', 'Toluene', 'Xylene', 'AQI',
       'AQI_Bucket'],
      dtype='object')
Mean Squared Error (MSE): 2383.59
R² Score: 0.46

Model Coefficients:
NO2: 3.8161
SO2: 2.2891
PM10: 42.0649

Prediction Results Sample:
        Actual   Predicted
325     46.52   75.929065
1385   178.87  206.001209
2148    77.71   88.398988
746    147.97  149.945822
299     84.03   84.018508
```

In [ ]:

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